

PMorph: A Task-Specialized Robot Manipulator Morphology Optimizer

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Abstract—This paper seeks to address the overactuation, energy-inefficiency, and lack of task specialization seen in manufacturing from the current suite of generalized robotic manipulators. The approach involves specifying a task in a MuJoCo simulated environment and then using a PPO agent to optimize the morphology for that environment given a reward function. The morphological elements to be optimized include the number of links, the lengths of each link, and the type of joints involved. The reward function is tuned to weigh energy cost, end effector movement, end effector accuracy, manipulability measures, number of links, and task completion. PPO is ideal for this task because it can efficiently optimize over the continuous action space that is morphology, resulting in a robot that is more simple, efficient, and capable than those commonly used in industry today.

I. INTRODUCTION

Imagine a factory where every employee has to be trained on and work every job in the plant. Imagine how inefficient that would be. Every employee would have general knowledge and ability, but no one would have a chance to specialize and become very good at any particular job. This is exactly the state of robotics in manufacturing today where high costs and lack of tailorability are some of the most cited problems for the adoption of robotics [1][2]. To make matters worse [3] and [4] show that most manipulators are overactuated, less cost-efficient, and less effective for most tasks they are used in today.

This paper seeks to address these issues by using a PPO agent to optimize the morphology of a robotic manipulator to the particular task it is being used for. This ensures that the robot is as simple, efficient, and capable as possible for its particular job while still allowing the user to specify the level of robustness by tuning the reward function.

PPO was chosen for its ability to explore the continuous action space that is morphology and it has been shown to do so efficiently and robustly [5]. The core method is to use `stable.baselines3` PPO implementation to iterate on the design parameters (number of links, joint types (X, Y, Z hinge, and Z actuation), and link lengths), use MuJoCo to simulate that iteration on a simple task using RRTConnect and damped least squares (DLS) inverse kinematics (IK) for the path planning, feed the results (success, efficiency, and manipulability index) back into the PPO model. RRTConnect was chosen for simplicity and it has been shown to be faster than other methods [6] which enables faster and more robust training. DLS IK was chosen for a similar reason, it has

been shown to be robust especially in situations where the target position is out of reach [7] which is inevitable when exploring morphologies, some of which will be too small or otherwise poorly suited to the task as the agent learns.

The evaluation signals will be successful path planning, path length, energy cost, and manipulability index. The manipulability index is a measure of a manipulators ability to arbitrarily change the position and orientation of the end effector at a particular point, borrowed from [8]. The tasks will be kept simple but varied in ways to test different types of movements like moving near the base, moving from near to far, different elevations, more linear movements, and different obstacle constraints. One might imagine that a lot of pick and place tasks are mostly linear and so might largely be done by linear actuators more efficiently for example.

The research questions to be answered: Can PPO be used to generate more efficient morphologies? Can morphologies be generated for tasks that generalized manipulators cannot do or cannot perform well? Can more simple morphologies (less DOF) be generated that can perform the same tasks with similar levels of accuracy?

II. RELATED WORK

Most of the research in this area codesigns morphology and control of atomic, modular robots. [9] shows that PPO can be used to generate freeform morphologies optimized for locomotion tasks from a set of atomic modules. [10] shows that reinforcement learning can be used on a modular robot to co-optimize morphology and control and outperform optimized control on a fixed morphology. [11] also uses reinforcement learning to co-optimize with the addition of a pretrained foundation model to accelerate the training process and improve results.

Most pertinent to this paper, [12] shows that PPO and A3C can be used to generate optimal morphologies that improve performance of reconfigurable robots, [13] uses DDQN to co-optimize morphology and control for modular robotic manipulators, and [14] uses soft actor critic for co-optimization showing that the approach resulted in drastically less design prototypes. This background motivates the approach of this paper which is going to use simple controllers and similar reinforcement learning techniques to maximize morphological potential. FANUC 6-DOF series of manipulators are the most commonly used in industry today [15][16] and so their morphology will be used as a baseline for comparison. Given the same environment, task, and controllers which morphologies will perform better?

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III. PROBLEM FORMULATION

The task is formulated as a single-step reinforcement learning problem in a MuJoCo simulated environment. A morphology θ is chosen, run through the simulation, and then evaluated based on weighted reward function terms.

A. Morphology Parameterization

A candidate morphology is defined by the tuple

$$\theta = (n, l, j),$$

where

- $n \in \{2, \dots, 7\}$ is the number of links,
- $l = [l_1, \dots, l_n] \in [0.05, 1.2]^n$ are the link lengths (m),
- $j = [j_1, \dots, j_n] \in \{0, 1, 2, 3\}^n$ are the joint types (0: revolute about x, 1: about y, 2: about z, 3: slide in z).

The action space of the RL agent is the continuous box

$$\mathcal{A} = [0, 1]^{1+7+7},$$

from which n , l , and j are decoded.

B. Task Specification

The robot must transport objects between start poses $\mathbf{s}_i$ and goal poses $\mathbf{g}_i$ (position + unit quaternion) inside a cluttered workspace containing fixed obstacles (container walls and floor). We use the *container task* (see Fig. ??), which captures typical manufacturing constraints: narrow access, tight tolerances, and collision risks. Poses are corrupted with Gaussian noise ($\sigma = 0.05$) at test time to encourage robust designs.

C. Evaluation Pipeline

Given a morphology θ , we evaluate it via the deterministic procedure $\text{eval}(\theta)$:

- 1) Generate a MuJoCo XML model (dynamic link/joint assembly).
- 2) Solve inverse kinematics for all start and goal poses using robust damped least-squares (DLS) with collision-aware line search.
- 3) If any IK fails (error > 0.1 rad/m), return reward -100 .
- 4) Plan a collision-free path using RRT-Connect.
- 5) If no path is found, return reward $30 - 100 \times \text{IK error}$.
- 6) Otherwise, compute:
 - End-effector path length $p = \sum \|\mathbf{x}_{ee}^{t+1} - \mathbf{x}_{ee}^t\|$,
 - Manipulability $\mu = \min(\sigma(\mathbf{J}))$ at start and goal (Yoshikawa, 1985),
 - Energy $E = \sum \|\boldsymbol{\tau} \odot \mathbf{v}\| \cdot \Delta t$ along the path (inverse dynamics at constant velocity segments).

D. Reward Function

The scalar reward is

$$r(\theta) = \frac{1}{N} \sum_{i=1}^N \left[100 \cdot \mathbb{I}[\text{success}] - p_i - 100 \cdot (e_{\text{start},i} + e_{\text{goal},i}) + 10 \cdot (\mu_{\text{start},i} + \mu_{\text{goal},i}) - (n-2) - 0.01 \cdot E_i \right],$$

TABLE I
AN EXAMPLE OF A TABLE

where $N = 2$ is the number of start-goal pairs, success is a valid IK + RRT path, and the weights were tuned to balance the competing objectives (reachability, shortness, dexterity,

simplicity, efficiency). The final optimization problem is therefore

$$\theta^* = \arg \max_{\theta} \mathbb{E}_{\pi(\cdot|\phi)} [r(\theta)],$$

where π is a PPO policy with parameters ϕ that directly outputs morphologies.

This formulation directly encodes the industrial challenges outlined in Section I: overactuation (penalized via link count and energy), lack of specialization (task-specific reward), and energy inefficiency (explicit power term). By optimizing the morphology itself as the RL action, we obtain designs that are provably better matched to the task than any fixed industrial arm (e.g., the 7-DoF Panda baseline we compare against).

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Positioning Figures and Tables: Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation “Fig. 1”, even at the beginning of a sentence.

One	Two
Three	Four

We suggest that you use a text box to insert a graphic (which is ideally a 300 dpi TIFF or EPS file, with all fonts embedded) because, in an document, this method is somewhat more stable than directly inserting a picture.

Fig. 1. Inductance of oscillation winding on amorphous magnetic core versus DC bias magnetic field

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity "Magnetization", or "Magnetization, M", not just "M". If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write "Magnetization (A/m)" or "Magnetization A[m(1)]", not just "A/m". Do not label axes with a ratio of quantities and units. For example, write "Temperature (K)", not "Temperature/K."

V. CONCLUSIONS

A conclusion section is not required. Although a conclusion may review the main points of the paper, do not replicate the abstract as the conclusion. A conclusion might elaborate on the importance of the work or suggest applications and extensions.

APPENDIX

Appendixes should appear before the acknowledgment.

ACKNOWLEDGMENT

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression, "One of us (R. B. G.) thanks . . ." Instead, try "R. B. G. thanks". Put sponsor acknowledgments in the unnumbered footnote on the first page.

References are important to the reader; therefore, each citation must be complete and correct. If at all possible, references should be commonly available publications.

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